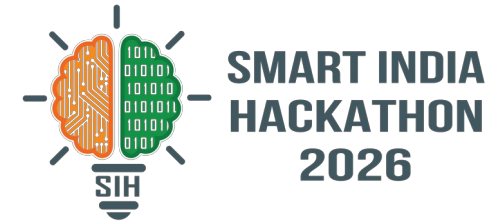


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TITLE PAGE

Problem Statement ID: SIH26183

Problem Statement Title

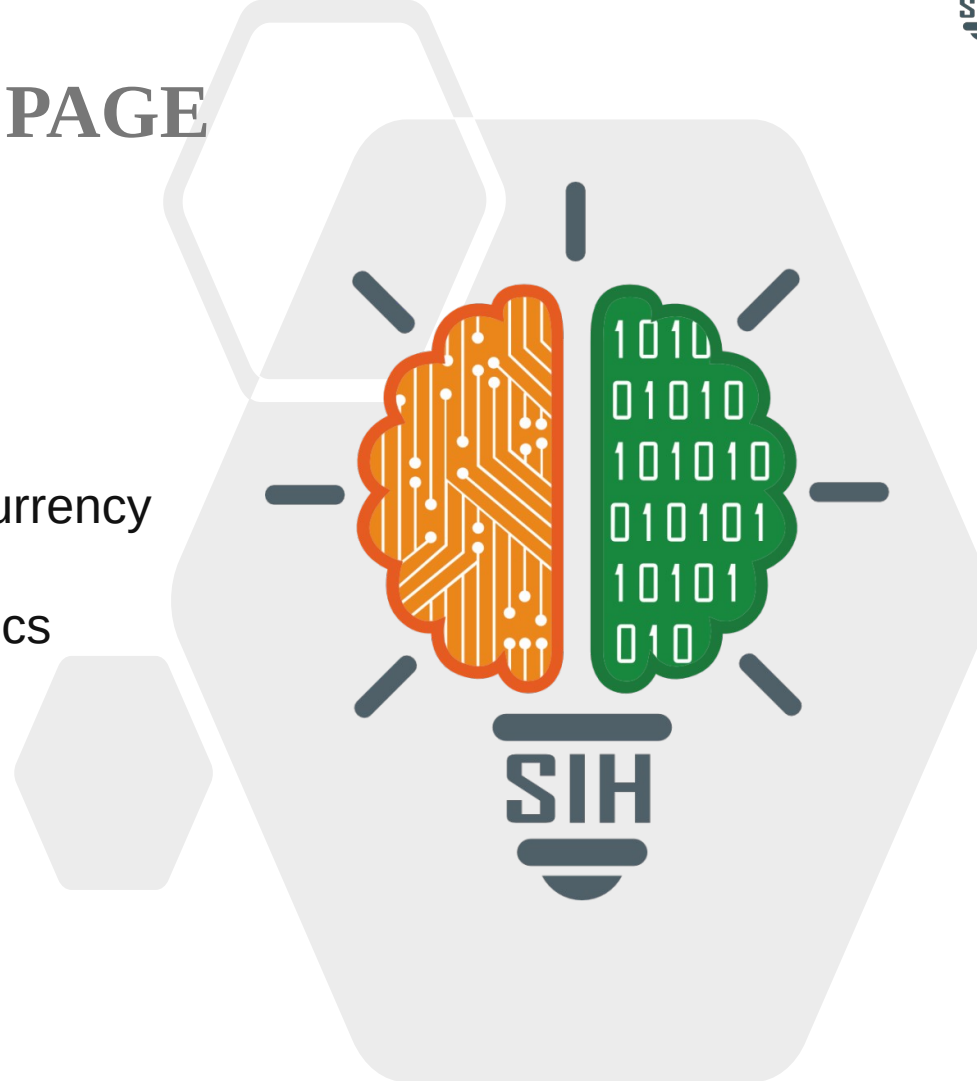
Real-Time Identification of Fraud-Linked Cryptocurrency Exchanges from Victim-Reported Suspect Wallet Addresses through Automated Blockchain Analytics

Theme: Blockchain & Cybersecurity

PS Category: Software

Team ID: [ENTER REGISTERED TEAM ID]

**Team Name (Registered on portal):
COLD COFFEE**



Proposed Solution (Describe your Idea/Solution/Prototype)

Detailed explanation of the proposed solution

- A reported wallet becomes an actionable receiving-VASP lead, with a reviewable fund trail.
- AEGIS will trace intermediary hops and identify the first supported receiving VASP on each path.
- Each finding will include transactions, amounts, timestamps, label sources and next actions.

How it addresses the problem

- Continuous monitoring will alert investigators when funds move or an actionable destination appears.
- A case-linked report will support preservation requests and coordination with the receiving VASP.

Innovation and uniqueness of the solution

- An evidence-linked case package will connect attribution reasoning directly to investigative action.
- New transactions or revised labels will update findings while preserving the earlier analysis.

Technologies to be used

React + FastAPI, queued workers, a transaction index and versioned VASP-label storage.

Methodology and process for implementation

- Trace time-consistent outgoing paths. Track spend links on Bitcoin and transfer events on EVM/TRON.
- Match sourced labels, cautiously expand clusters and rank receiving candidates by evidence strength.
- Correlate supported bridge events. Explainable rules and evaluated ML will flag laundering patterns.



Cross-chain correlation where supported; uncertainty remains explicit

Ambiguous ownership, mixing and unsupported bridges will remain explicit tracing gaps.

Analysis of the feasibility of the idea

- Label supply: exchange disclosures and reviewed public labels, supplemented by authorized VASP data.
- Delivery: validate Bitcoin attribution first, then monitoring, EVM/TRON and supported bridge adapters.

Potential challenges and risks

- Incomplete labels and ambiguous flows can produce missed matches or false attribution.
- Provider limits, sensitive case data and integration access constrain agency deployment.

Strategies for overcoming these challenges

- Index incrementally, cache queries and version every label with its source and review date.
- Test on held-out labelled paths and hard negatives. Measure precision, coverage and alert latency.
- Protect cases with role-based access and audit logs. Use approved NCRP/SAHYOG interfaces.

Potential impact on the target audience

- Investigators will receive a prioritized VASP lead with the evidence needed to review it.
- Case teams will share a reproducible report and receive alerts as the fund trail changes.

Illustrative investigation outcome

Reported Wallet A pays intermediary B. B later pays an address labelled to VASP X.

Finding: VASP X candidate, two-hop path, transaction IDs, amounts and label provenance.

Next action: investigator reviews the finding and prepares a VASP preservation request.

Illustration only. VASP X is fictional, and the path is not a demonstrated fraud case.

Benefits of the solution (social, economic, environmental, etc.)

- Social: help investigators act on suspected fraud proceeds while retaining human review.
- Economic: reduce repeated tracing work through shared indexing and reusable case reports.
- Operational: support consistent VASP coordination with a documented, auditable fund trail.

Details / Links of the reference and research work

Address clustering	Meiklejohn et al. (2013), A Fistful of Bitcoins: Characterizing Payments Among Men with No Names. ACM IMC paper
ML-assisted analysis	Weber et al. (2019), Anti-Money Laundering in Bitcoin: Experimenting with Graph Convolutional Networks for Financial Forensics. arXiv:1908.02591
Blockchain data	Blockstream Esplora: address and transaction API documentation. Official API documentation
Attribution sources	Binance, Our Commitment to Transparency (2022): published wallet disclosures. Historical disclosure; not a deposit-account registry
AML context	FATF (2021), Updated Guidance for a Risk-Based Approach to Virtual Assets and VASPs. FATF guidance
Evidence handling	NIST SP 800-86 (2006), Guide to Integrating Forensic Techniques into Incident Response. NIST guidance; not a guarantee of legal admissibility